BRIO Research Paper: Phase 3 Complete Learning Notes

IMU Preintegration, Bearing–Doppler Radar, Robust Losses, Barometry, and Zero-Velocity Tracking

Study notes based on Girod *et al.* (2024)

Key idea

Goal of Phase 3: Understand what each sensor contributes, how each measurement becomes a factor/residual, and how the factor graph and optimizer estimate the drone's position, velocity, orientation, and IMU biases. This document explains Sections IV-D–IV-G of the BRIO paper in beginner-friendly language; simple numerical examples are illustrative, not reported experimental results.

1 Recap: State Estimation from Phases 1 and 2

The state at radar time index i is

$$\boxed{x_i = \left(R^i_{\mathrm{IB}},\ {}^{\mathrm{I}}t^i_{\mathrm{IB}},\ {}^{\mathrm{I}}v^i_{\mathrm{IB}},\ {}^{\mathrm{B}}b^i_g,\ {}^{\mathrm{B}}b^i_a\right)}. \tag{3}$$

Component	Meaning
R^i_{IB}	Orientation of body frame relative to inertial frame (3D rotation matrix).
${}^{\mathrm{I}}t^i_{\mathrm{IB}}$	Drone/IMU position in the inertial frame (3 values).
${}^{\mathrm{I}}v^i_{\mathrm{IB}}$	Drone/IMU velocity in the inertial frame (3 values).
${}^{\mathrm{B}}b^i_g$	Gyroscope bias (3 values).
${}^{\mathrm{B}}b^i_a$	Accelerometer bias (3 values).

The orientation has three degrees of freedom, so the complete state has $3+3+3+3+3 = 15$ degrees of freedom. The three frames are I (fixed inertial/world), B (body/IMU), and R (radar). Sensors do not usually provide the full state directly: the estimator infers it from multiple measurements.

2 Overview of the Phase 3 Components

Component	Purpose
IMU preintegration and IMU factor	Combine high-rate acceleration/rotation measurements to constrain relative motion between successive states.
Bearing–Doppler radar factor	Compare predicted and measured radial velocity for each radar detection.
Robust loss for radar	Reduce influence of moving targets and ghost reflections; BRIO selects Welsch.
Barometer factor	Compare estimated vertical position with pressure-derived relative height; BRIO uses Fair loss.
Zero-velocity detection tracking	Link repeated stationary radar detections to poses and shared landmarks, especially before takeoff and after landing.

3 Topic 4: IMU Preintegration (Section IV-G, Eq. 15, p. 5)

3.1 What does an IMU measure?

An IMU has an **accelerometer** and a **gyroscope**.

- An accelerometer measures *specific force*, which is acceleration-related but also reflects gravity; using it to predict motion requires accounting for orientation, gravity, and accelerometer bias.
- A gyroscope measures *angular velocity*, which tells us the rate of change of orientation; it also has bias.

One sample may be represented by

$$z_{\text{IMU},k} = \begin{bmatrix} a_x & a_y & a_z & \omega_x & \omega_y & \omega_z \end{bmatrix}_k^T. \tag{1}$$

For example, an angular rate of $30°/\text{s}$ maintained for $2\,\text{s}$ gives $\Delta\theta = 30 \times 2 = 60°$ in a simple one-axis case. Likewise, a constant gravity-corrected acceleration of $2\,\text{m/s}^2$ for $1\,\text{s}$ changes velocity by $2\,\text{m/s}$.

3.2 How acceleration predicts velocity and position

For a *one-dimensional, constant-acceleration* illustrative case,

$$v_{k+1} = v_k + a\Delta t, \tag{2}$$

$$p_{k+1} = p_k + v_k\Delta t + \tfrac{1}{2}a\Delta t^2. \tag{3}$$

If $p_0 = 0$, $v_0 = 0$, $a = 2\,\text{m/s}^2$, and $\Delta t = 1\,\text{s}$, then $v_1 = 2\,\text{m/s}$ and $p_1 = 1\,\text{m}$. In 3D, acceleration must be rotated to the correct frame and gravity must be incorporated.

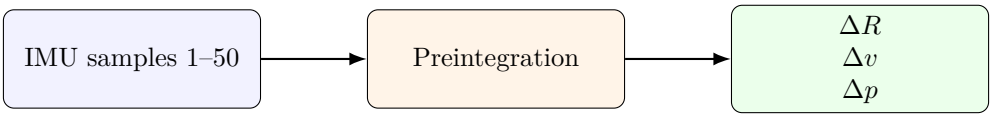
3.3 Why do we receive many IMU samples between radar frames?

The experimental configuration reports accelerometer sampling at $400\,\text{Hz}$ and radar triggering at approximately $8\,\text{Hz}$. Hence the nominal radar interval is

$$\Delta t_{\text{radar}} = \frac{1}{8} = 0.125\,\text{s}, \qquad 400 \times 0.125 \approx \boxed{\text{50 accelerometer samples}}. \tag{4}$$

There are also gyroscope measurements synchronized with the IMU stream. Creating a distinct optimized state for every IMU sample would make the factor graph much larger. Instead, BRIO creates a new state for each radar frame and **preintegrates** IMU measurements in between.

3.4 Meaning of ΔR, Δv, and Δp



- $\boxed{\Delta R}$: accumulated relative orientation change (a 3D rotation, not just an angle).
- $\boxed{\Delta v}$: accumulated relative velocity contribution from accelerometer readings.
- $\boxed{\Delta p}$: accumulated relative position contribution from accelerometer readings.

Important distinction: The preintegrated Δv and Δp are *not necessarily* the simple world-frame differences $v_{i+1} - v_i$ and $p_{i+1} - p_i$. The full state-to-state prediction also needs the starting orientation, starting velocity, gravity, and IMU bias corrections.

Worked example

Assume for demonstration that over one radar interval of 0.125 s, rotation is a constant 10°/s around one axis and gravity-corrected acceleration is a constant $2\,\mathrm{m/s^2}$ along a fixed direction. Then

$$\Delta\theta = 10(0.125) = \boxed{1.25^\circ},$$
$$\Delta v \approx 2(0.125) = \boxed{0.25\,\mathrm{m/s}},$$
$$\Delta p \approx \tfrac{1}{2}(2)(0.125)^2 = \boxed{0.015625\,\mathrm{m}}.$$

The Δp figure is the *acceleration contribution*, not total displacement. If initial velocity is 1 m/s along the same axis, total displacement is approximately $1(0.125) + 0.015625 = 0.140625\,\mathrm{m}$. In a real rotating 3D drone, the preintegration calculation properly accounts for relative rotations and biases.

3.5 Why IMU-only estimates drift

Small gyroscope and accelerometer errors are integrated over time. An acceleration bias of $0.02\,\mathrm{m/s^2}$, if uncorrected, creates a velocity error that grows with time; integrating velocity again creates position drift. Orientation errors can also cause incorrect gravity compensation, amplifying drift. IMU factors therefore benefit from radar and barometer constraints.

3.6 What is the IMU factor and its output?

The paper's Eq. (15) stacks five residual components:

$$\boxed{r_I^{i,i+1} = \begin{bmatrix} r_{\Delta R}^{i,i+1} \\ r_{\Delta t}^{i,i+1} \\ r_{\Delta v}^{i,i+1} \\ r_{\Delta b_g}^{i,i+1} \\ r_{\Delta b_a}^{i,i+1} \end{bmatrix}}. \tag{15}$$

These terms compare the motion implied by the two estimated states with the preintegrated IMU measurement, and also constrain changes in the gyroscope and accelerometer biases. The IMU factor **connects state x_i to state x_{i+1}**. The optimizer reduces its covariance-weighted residual along with other factors.

4 Topic 5: Radar Bearing–Doppler Factor (Section IV-D, Eqs. 8–10, p. 3)

4.1 What does radar provide?

A radar detection contains target direction (direction of arrival, DoA), Doppler/radial velocity, range, signal-to-noise ratio, and noise information. For the *bearing–Doppler factor*, BRIO uses direction and Doppler velocity. Range is also relevant to the separate zero-velocity tracking factor.

4.2 What is radial (Doppler) velocity?

Doppler reports motion *along the line of sight to a target*; a single detection does *not* reveal complete 3D velocity. For a stationary target directly ahead, a drone traveling toward it at 2 m/s gives radial speed magnitude 2 m/s, with sign determined by the radar convention.

4.3 What is direction of arrival?

${}^{\mathrm{R}}e_T \in \mathbb{R}^3$ is a unit vector pointing from radar to target, written in radar coordinates. Examples are $(1,0,0)^T$, $(0,1,0)^T$, and $(0,0,1)^T$. The radar frame and IMU/body frame may differ; their fixed calibration must be used to transform vectors consistently.

4.4 Why at least three linearly independent directions?

The instantaneous unknown velocity has three components v_x, v_y, v_z. Each stationary-target Doppler measurement gives one scalar projection equation. Three independent directions make the associated 3×3 direction matrix full rank, so the 3D radar velocity is identifiable from Doppler alone. Three identical or coplanar dependent directions are insufficient. The complete BRIO estimator also has IMU and other constraints; this statement concerns Doppler-only recovery of instantaneous 3D velocity.

4.5 How the Doppler residual is formed

The paper defines

$$\boxed{r_D^{i,m} = -({}^{\mathrm{R}}v_{\mathrm{IR}}^i)^T \, {}^{\mathrm{R}}e_T^{i,m} - v_T^{i,m}.} \tag{8}$$

Here i indexes a radar frame and m indexes a detection. The quantity $-({}^{\mathrm{R}}v_{\mathrm{IR}}^i)^T {}^{\mathrm{R}}e_T^{i,m}$ is the **predicted Doppler** for a static target using the paper's sign convention, and $v_T^{i,m}$ is the **measured Doppler**. The residual is predicted minus measured.

Worked example

Let radar velocity be ${}^{\mathrm{R}}v_{\mathrm{IR}} = (2,0,0)^T\,\mathrm{m/s}$ and target direction ${}^{\mathrm{R}}e_T = (1,0,0)^T$. Then predicted Doppler is $-2\,\mathrm{m/s}$. If measured Doppler is $-1.8\,\mathrm{m/s}$,

$$\boxed{r_D = -2 - (-1.8) = -0.2\,\mathrm{m/s}.}$$

The factor tells the optimizer that the current state prediction disagrees with this detection by $0.2\,\mathrm{m/s}$.

4.6 Why the radar's mounting offset matters (Eq. 10)

The radar may be located some distance from the IMU. When the drone rotates, this offset creates an additional radar linear velocity. The paper computes

$$\boxed{{}^{\mathrm{I}}v_{\mathrm{IR}}^i = {}^{\mathrm{I}}v_{\mathrm{IB}}^i + R_{\mathrm{IB}}^i \left[\left({}^{\mathrm{B}}\omega_{\mathrm{IB}}^i - {}^{\mathrm{B}}b_g^i \right) \times {}^{\mathrm{B}}t_{\mathrm{BR}} \right].} \tag{10}$$

In words: *radar velocity = body/IMU velocity + velocity caused by rotation around the IMU-to-radar lever arm.* Equation (9) applies the coordinate transformation and the same Doppler projection in radar coordinates.

5 Topic 6: Robust Loss Functions (Section IV-D, Fig. 4, pp. 3–4)

5.1 What are a residual and an outlier?

A residual compares predicted and measured data. An outlier is a measurement especially inconsistent with the assumed model. The radar factor assumes detected targets are static, but a moving car, pedestrian, streetcar, or multipath ghost may violate that assumption.

5.2 Why quadratic loss can be a problem

Ordinary quadratic loss is $L(r) = r^2$. For illustration, a residual $r = 0.2$ yields 0.04, while $r = 8$ yields 64. One extreme error can overwhelm many ordinary measurements and pull the optimized state toward an incorrect solution.

5.3 What is a robust loss?

A robust loss changes how residual size translates into optimization cost, giving large outliers less influence than ordinary squared error. It does not necessarily delete a measurement. BRIO compares quadratic, Huber, Fair, Cauchy, and Welsch losses in Figure 4.

5.4 Why does the paper choose Welsch for radar?

In the paper's streetcar experiment, the drone hovers while a moving streetcar produces inconsistent Doppler returns. A correct velocity estimate should stay near zero. Figure 4 shows Welsch yields the steadiest hover estimate among the tested losses; the authors choose Welsch for radar factors.

Key idea

For Doppler: radar factor + Welsch robust loss. Large radar residuals are strongly down-weighted, so moving objects and ghost detections have less impact on the optimized state.

The paper normalizes radar residuals using Doppler standard deviation σ_D and also weights residuals by covariance. The loss controls influence of outliers; covariance encodes measurement uncertainty. These have different roles.

6 Topic 7: Barometer Factor (Section IV-E, Eqs. 11–12, pp. 4–5)

6.1 What does a barometer measure?

A barometer measures atmospheric pressure p. Pressure generally decreases with increasing altitude. BRIO converts pressure to a height-like value z_p^i using the atmosphere model in Eq. (12). It then uses *differential* height relative to the initial barometer reading, rather than assuming absolute altitude is known perfectly.

6.2 Why barometry is needed

IMU integration, imperfect radar direction estimates, calibration errors, and bias can cause drift in estimated vertical position. Pressure-derived relative height adds a separate vertical constraint.

6.3 Barometer residual (Eq. 11)

$$\boxed{r_B^i = ({}^{\mathrm{I}}e_z)^T {}^{\mathrm{I}}t_{\mathrm{IB}}^i + z_p^0 - z_p^i.} \tag{11}$$

The term $({}^{\mathrm{I}}e_z)^T {}^{\mathrm{I}}t_{\mathrm{IB}}^i$ extracts estimated vertical position; $z_p^i - z_p^0$ is the relative height suggested by barometry. Thus

$$\boxed{r_B = \text{estimated relative height} - \text{barometer-derived relative height.}}$$

Worked example

Suppose the estimated relative height is 12 m, the starting pressure-derived height is $z_p^0 = 100$ m, and the current pressure-derived height is $z_p^i = 110$ m. Then

$$\boxed{r_B = 12 + 100 - 110 = 2\,\text{m}.}$$

The estimate claims a 12 m climb, while barometry suggests 10 m. The optimizer uses this 2 m disagreement as a vertical constraint.

6.4 Why Fair loss for barometry?

Propeller downwash and ground effect during touchdown can disturb measured pressure without a corresponding true height change. Figure 5 (page 5) shows that Fair robust loss helps keep the estimated vertical velocity smooth during such pressure disturbances. Thus BRIO uses

$$\boxed{\text{barometer factor} \;+\; \text{Fair robust loss}.}$$

The paper fixes the initial pressure-height offset from the first measurement rather than jointly estimating a freely varying barometer bias. Rapid ambient pressure changes remain a limitation.

7 Topic 8: Zero-Velocity Detection Tracking (Section IV-F, Eqs. 13–14, p. 4)

7.1 Problem: Doppler information can be sparse at standstill

Imagine a drone stationary on the ground and a wall 5 m in front of it. Drone velocity and wall velocity are both zero, so ideal wall Doppler is 0 m/s. Several stationary targets share this zero Doppler value. The CFAR radar detector can struggle to separate them in Doppler space (some remain distinguishable by range), leaving only one or two strong detections. Such few Doppler directions may not determine 3D drone velocity.

7.2 Why zero Doppler from one wall is not enough

A single wall directly ahead tells us there is no motion *toward or away from that wall*; the drone might still slide sideways parallel to it. So zero radial velocity is not equivalent to a proven zero 3D velocity.

7.3 Solution: recognize the same stationary target across frames

The paper's idea is to match repeated radar returns believed to come from the same stationary object. Unlike a Doppler-only factor, tracking uses the *observed target position relative to the radar* across frames. For example,

$${}^{\mathrm{R}}t_{\mathrm{R}T}^{1,m} = \begin{bmatrix}3\\2\\1\end{bmatrix}\text{m}, \qquad {}^{\mathrm{R}}t_{\mathrm{R}T}^{2,n} = \begin{bmatrix}3\\2\\1\end{bmatrix}\text{m}.$$

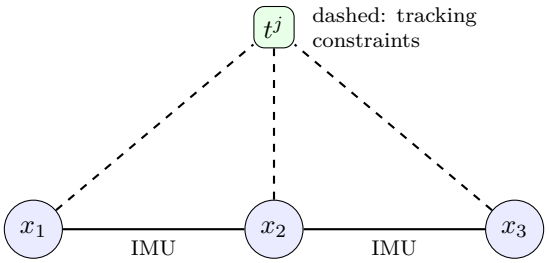
If a stationary reflector is seen at the same relative position and direction, this is consistent with the drone staying still. A single identical target observation alone does not establish all six pose degrees of freedom; it provides a useful constraint when combined with other factors.

7.4 What exactly is "tracking" (Eq. 13)?

Tracking means declaring that a detection in frame i corresponds to a detection in another frame. Section IV-F describes association when the detections have identical reported target position, zero Doppler, signal-to-noise ratio, and noise information (Eq. 13). Tracks are eventually discarded if unobserved for a set number of frames. The tracked target is represented by a *landmark* with inertial/world position t^j. BRIO limits this tracking to standstill/static scenes rather than general flight mapping.

7.5 How a landmark connects states

Suppose one reflector is observed in three successive radar frames. The factor graph contains drone states x_1, x_2, x_3 and one shared landmark t^j.



Each observation ties an estimated pose to the *same* stationary target. If poses drift while measurements remain fixed, the predicted target positions begin to disagree with radar detections.

7.6 Tracking residual (Eq. 14)

In the paper's notation, the predicted target coordinate in the radar frame is obtained from the world landmark and the estimated body/radar pose. The residual is

$$\boxed{r_T^{i,j} = (R_{\mathrm{IB}}^i R_{\mathrm{BR}})^T \left(t^j - {}^{\mathrm{I}}t_{\mathrm{IB}}^i - R_{\mathrm{IB}}^i \, {}^{\mathrm{B}}t_{\mathrm{BR}} \right) - {}^{\mathrm{R}}t_{\mathrm{R}T}^{i,m}.} \tag{14}$$

Symbol	Meaning
$r_T^{i,j}$	Tracked-target position residual for state i and landmark j.
t^j	Estimated fixed landmark position in inertial coordinates.
$R_{\mathrm{IB}}^i, R_{\mathrm{BR}}$	Body orientation and known body-to-radar calibration rotation.
${}^{\mathrm{I}}t_{\mathrm{IB}}^i$	Estimated drone/IMU world position.
${}^{\mathrm{B}}t_{\mathrm{BR}}$	IMU-to-radar mounting translation, expressed in body coordinates.
${}^{\mathrm{R}}t_{\mathrm{R}T}^{i,m}$	Measured target position relative to radar.

In plain language,

$$\boxed{\text{Tracking residual} = \text{predicted target position} - \text{measured target position.}}$$

7.7 Detailed numerical drift example: the wall at 5 metres

1. The drone is *actually stationary* at 0 m, and the wall is at 5 m in a simple 1D world.
2. At time t_1, radar sees the wall 5 m away; later at t_2 it again sees the same wall 5 m away.
3. Because IMU estimates drift, suppose the drone position estimate at t_2 is incorrectly 0.5 m toward the wall.
4. Given the tracked wall landmark at 5 m, predicted radar distance is $5 - 0.5 = 4.5$ m.

5. The actual radar measurement is 5.0 m. Hence the tracking residual is

$$\boxed{r_T = 4.5 - 5.0 = -0.5\,\text{m}.}$$

6. The factor graph optimizer uses this disagreement, together with IMU and other measurements, to adjust the state and landmark estimates. This helps resist false drift.

Key idea

Answer to our discussion question: The repeated radar wall observation reveals the inconsistency: the estimated drone motion predicts the wall should appear closer, but the measured wall position has not changed. Tracking does not magically declare every IMU estimate wrong; the optimizer weighs both constraints and adjusts their jointly estimated variables.

7.8 When is this useful?

Before takeoff and after landing, a stationary drone can still appear to move due to IMU noise and bias. A repeated stable radar reflection constrains pose even when Doppler detection directions are insufficient. The paper notes a limitation: if no repeatable CFAR detection exists, tracking cannot provide this constraint and the estimator can still drift.

8 From Measurements to Optimized State: Complete Factor Graph Flow

8.1 Each measurement produces a constraint

Factor	Input and relationship	Main output
IMU	Accelerations/angular rates between i and $i+1$; bias-aware preintegration	Residual $r_I^{i,i+1}$ linking x_i, x_{i+1}
Doppler	Direction and Doppler of each radar detection	Residual $r_D^{i,m}$ constraining velocity; Welsch loss
Barometer	Pressure converted to relative height	Residual r_B^i constraining altitude; Fair loss
Tracking	Same stationary radar target observed repeatedly	Residual $r_T^{i,j}$ linking pose and landmark
Prior	Initial orientation, position, velocity, biases	Residual r_P^0 anchoring initialization

Remember: Factors are *constraints/functions*, not merely sensor values. The factor graph stores their connections to unknown states; the optimizer solves the joint problem. Robust loss is applied when evaluating appropriate factor costs, not as an independent physical sensor.

8.2 Optimization idea (Section IV-C, Eq. 7)

A simplified conceptual total cost is

$$J(X,T) = \|r_P^0\|^2_{\Sigma_P} + \sum_i \|r_I^{i,i+1}\|^2_{\Sigma_I} + \sum_{i,j} \|r_T^{i,j}\|^2_{\Sigma_T} \tag{5}$$

$$+ \sum_{i,m} \text{robust-Welsch Doppler cost}(r_D^{i,m}, \Sigma_D, \sigma_D) \tag{6}$$

$$+ \sum_i \text{robust-Fair barometer cost}(r_B^i, \Sigma_B, \sigma_B), \tag{7}$$

where $\|r\|_{\Sigma}^2 = r^T \Sigma^{-1} r$ is the squared Mahalanobis distance. The paper's Eq. (7) specifies how the normalized robust losses and uncertainty-weighted residual terms are combined. Lower uncertainty generally means a stronger constraint; larger uncertainty generally means a weaker constraint.

8.3 Workflow diagram

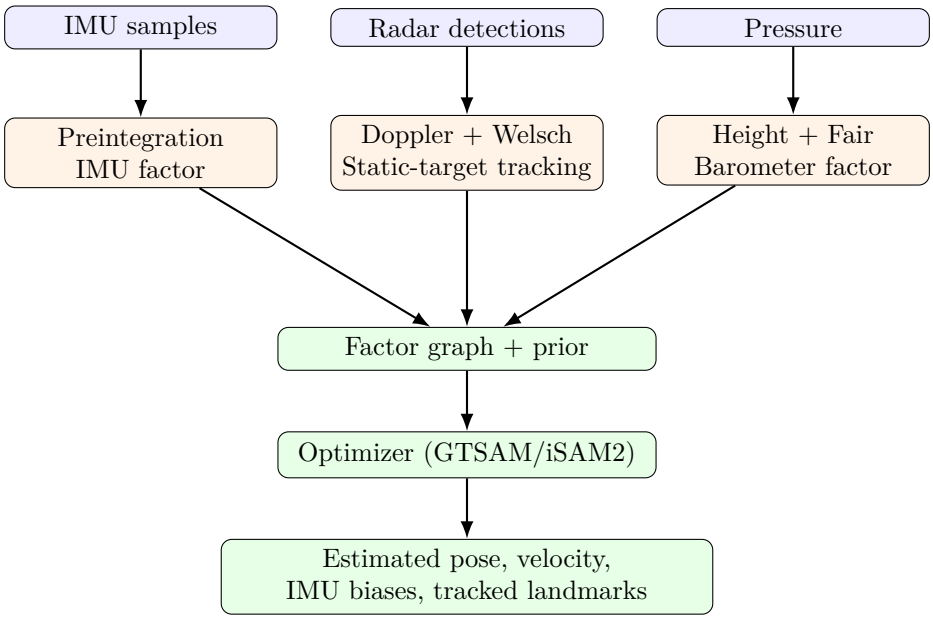


Diagram clarification: Doppler constraints and stationary-target tracking are two distinct uses of radar detections; tracking does not follow from Welsch loss. The drawing groups them visually only because both begin with radar data.

9 How This Connects to Reproducing the Paper

The paper's Section V states that BRIO uses GTSAM with the iSAM2 solver, with an approximately 10 s optimization data window. For a reproduction, you should be able to identify (1) incoming sensor values and timestamps, (2) calibration and bias/uncertainty information, (3) factor construction, (4) state and landmark variables, and (5) the optimized state returned for navigation. The paper's source code is linked at `https://github.com/ethz-asl/rio`.

10 Quick Revision and Oral-Exam Questions

1. **Why preintegrate IMU samples?** Many high-rate IMU measurements are combined into one relative-motion factor between radar-time states, reducing optimization burden.
2. **Are $\Delta R, \Delta v, \Delta p$ final optimized states?** No. They are preintegrated relative-motion quantities used by the IMU factor.
3. **What does a Doppler factor compare?** Predicted stationary-target line-of-sight velocity versus measured Doppler.
4. **Why three independent detections?** To recover an arbitrary 3D instantaneous velocity from scalar Doppler projections alone.
5. **Why Welsch?** To strongly suppress moving-target and ghost-return Doppler outliers.
6. **What does the barometer constrain?** Relative height/vertical position; Fair loss handles occasional pressure disturbances.

7. **Why tracking at zero velocity?** Few separable radar Doppler targets can leave motion poorly constrained. Repeated stationary target positions add pose constraints.
8. **What does the optimizer receive?** A factor graph containing residual functions, uncertainty weights, robust losses where specified, and state/landmark variables.
9. **What comes out?** An optimized estimate of position, orientation, velocity, IMU biases, and any actively tracked landmarks.

Paper Reference

R. Girod, M. Hauswirth, P. Pfreundschuh, M. Biasio, and R. Siegwart, "A robust baro-radar-inertial odometry m-estimator for multicopter navigation in cities and forests," arXiv:2408.05764v1, 2024. This document is an explanatory learning companion to the uploaded paper, not a verbatim reproduction. Page and equation references correspond to that version.